

Chronic Allograft Injury

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DEFINITIONS AND EPIDEMIOLOGY

The results after kidney transplantation have steadily improved over the past 30 years with a 1-year overall graft survival rate of about 94% and acute rejection rates within the first year after transplantation of 12%.¹ Loss of an allograft from acute rejection is now rare and is predominantly found in patients who are presensitized against human leukocyte antigens (HLAs).² Today, most allografts are lost by chronic antibody-mediated rejection that is a consequence of (de novo) development of donor-specific HLA antibodies (DSAs) but also non-HLA antibodies.^{3,4,5} Every year, about 4% of kidney grafts are lost (Fig. 112.1). Only 2 decades ago, it was believed that nonimmunologic causes were the main factors leading to chronic allograft injury and graft loss.⁶ Today, HLA alloantibodies are believed to be the main culprit, accounting for more than half of graft losses in the long term.⁷ With the acknowledgment of alloantibodies as main factors that contribute to graft failure, the term *chronic allograft nephropathy* has been replaced by more specific terminology in the Banff classification of kidney allograft pathology (Table 112.1).^{8,9}

Although immunologic causes are mainly responsible for late allograft injury and graft failure, nonimmunologic causes still represent significant risk factors for (late) graft injury and graft loss. The donor graft may present with significant preexisting disease limiting long-term graft survival.¹⁰ In addition, chronic interstitial fibrosis and tubular atrophy (IFTA) may result from early damage to the allograft such as ischemia/reperfusion injury or early acute rejection. Calcineurin-inhibitor (CNI) nephrotoxicity, recurrent or de novo glomerular disease, BK virus infection, and IFTA originating from other causes further contribute to graft damage and graft loss over the long term.

Although some allografts are lost because of one specific cause, others accumulate damage from several different causes with gradual loss of functioning nephrons. Main contributors to late graft injury and late graft loss are summarized in Box 112.1.

PATHOGENESIS: NONIMMUNOLOGIC FACTORS

Donor Age, Donor Sex, and Donor-Recipient Size Mismatching

Long-term graft survival is reduced in kidneys from older donors, an effect that is more pronounced in deceased compared with living kidney transplant recipients.¹¹ Impaired graft survival is attributed to a differential response of the older organ to injury, an impaired capacity to withstand stress, a limited ability to repair structural damage, or amplification of external injury because of preexisting structural abnormalities.¹²

Reduced nephron mass resulting in glomerular hyperfiltration and hypertension with accelerated senescence is also a postulated

mechanism for a progressive decline in kidney graft function.¹³ The effect of kidney size mismatching is thought to be related to insufficient numbers of nephrons within the donor kidney, which, in turn, leads to compensatory hyperfiltration.¹⁴ Although some experimental data support this hypothesis, the extent to which an inadequate number of nephrons contributes to chronic injury in clinical transplantation is unknown.

Female donor sex has a negative impact on kidney graft survival, with reduced survival when female grafts are transplanted into male recipients (risk ratio [RR], 1.22). This may be the result of reduced nephron mass of female donor kidneys (nephron underdosing). Other mechanisms may include possible differences in the immunogenicity of male and female grafts.^{15,16}

Ischemia/Reperfusion Injury and Delayed Graft Function

Graft survival is lower in recipients with longer ischemia times, and longer cold ischemia time represents a risk factor for the development of IFTA at 6 months after transplantation.¹⁷ Ischemia-reperfusion injury also may trigger immune-mediated injury because ischemia and oxidative injury resulting from reperfusion are associated with activation of the adaptive immune response, antigen-presenting cells and Toll-like receptors, and release of proinflammatory cytokines—all of which can lead to acute rejection and subsequent IFTA.¹⁸ Pretreatment of the deceased organ donor with low-dose dopamine (aiming to reduce ischemia/reperfusion injury) reduced the need for dialysis early after kidney transplantation but failed to improve long-term kidney graft survival.^{19,20} Similarly, therapeutic hypothermia compared with normothermia of the deceased organ donor was associated with a lower incidence of delayed graft function in the kidney transplant recipient (28% vs. 39%).²¹ Another promising approach to reduce ischemia/reperfusion injury and improve early kidney graft function is the replacement of cold storage of kidneys by pulsatile machine perfusion during transportation. This strategy has been tested in clinical trials and is now routine practice in some centers.²²

Delayed graft function, commonly defined by the need for dialysis in the first week after transplantation, is a risk factor for IFTA.²³ There are also independent associations between delayed graft function and late graft failure and between delayed graft function and graft failure that is mediated by acute rejection. Major risk factors for delayed graft function are prolonged cold ischemia time, donor age older than 50 years, diabetes mellitus as a cause of end-stage kidney disease (ESKD), time on dialysis of more than 5 years, and the presence of HLA alloantibodies.²⁴ The risk for delayed graft function is reduced when the donor or recipient is of female sex or when trauma is the cause of donor death. Ischemia and oxidative injury resulting from reperfusion of an ischemic kidney may cause an upregulation of major histocompatibility complex (MHC) antigens or proinflammatory cytokines, predisposing to acute rejection.

Graft Survival for Recipients of a First Deceased Donor Kidney Transplant, by Year of Transplant

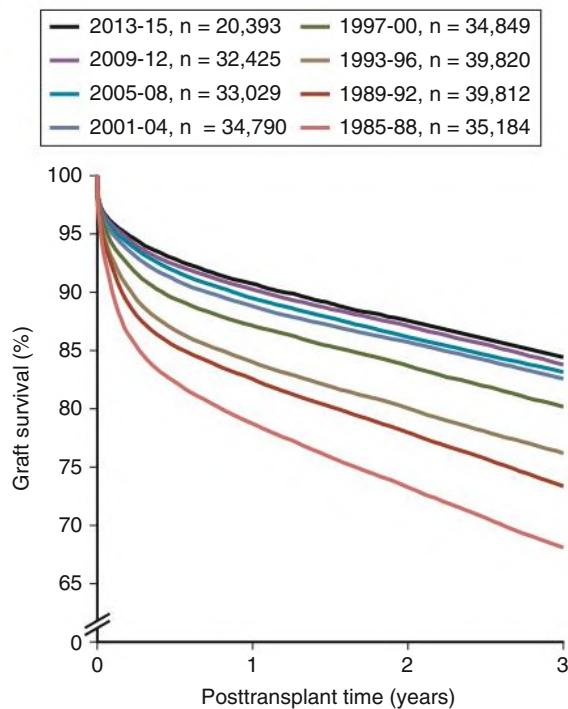


Fig. 112.1 Graft Survival for Recipients of a First Deceased Donor Kidney Transplant According to Year of Transplantation. Graft survival has steadily improved over the past years but mainly as a result of reduction of early graft loss. (Courtesy Prof. Gerhard Opelz, CTS Study.)

BK Virus Nephropathy

BK virus is an endemic polyomavirus that is usually acquired during childhood and persists in the urinary tract. Although it is asymptomatic in the immunocompetent host, it becomes activated in the immunocompromised patient. Viremia may be found in 10% to 30% of patients within the first 6 months after transplantation with biopsy-proven BK virus–associated nephropathy in 1% to 10% of patients. BK virus–associated nephropathy mimics the pattern of interstitial cellular rejection but may be distinguished in immunohistochemistry by positive staining for the SV40 antigen. The Banff working group identified two histologic markers predictive for graft function and graft survival. The Banff interstitial fibrosis (ci) score and a new score termed “the intrarenal polyomavirus load level (pvl).” The pvl score is based on the fraction of tubules with evidence of BK replication by light microscopy or immunohistochemical staining of epithelial cell nuclei for SV40 antigen. The pvl and ci scores were used to define the following three BK virus–associated nephropathy classes: pvl1 up to 1%, pvl2 greater than 1%, and less than 10%, and pvl3 of 10% or greater.²⁵ BK virus–associated nephropathy, if untreated, may lead to severe IFTA and graft loss in most patients. The mainstay of therapy is the reduction of immunosuppression.²⁶ The diagnosis and management of BK virus nephropathy are discussed further in [Chapter 110](#).

Calcineurin Inhibitor Toxicity

CNI nephrotoxicity affects all histologic compartments of the transplanted kidney. Although not specific for CNI toxicity, CNI lesions include medial arteriolar hyalinosis, striped interstitial fibrosis, global

glomerulosclerosis, and tubular microcalcification unrelated to other causes, such as tubular necrosis and hyperparathyroidism.^{27,28} CNI-induced arteriopathy is characterized by nodular hyaline deposits in the media of afferent arterioles sufficient to cause narrowing of the vascular lumen.^{29,30} It is attributed to eosinophilic transformation and vacuolization of smooth muscle cells with subsequent necrosis. Arteriolar hyalinosis is the most reliable diagnostic marker of CNI nephrotoxicity. Confirmation of the diagnosis is made by exclusion of other causes, such as donor hyalinosis (which can be detected on the implantation biopsy specimen), diabetes, and hypertensive nephrosclerosis. Striped fibrosis is subjectively defined by a dense stripe of cortical fibrosis and atrophic tubules adjacent to normal cortex and is traditionally regarded as pathognomonic of CNI nephrotoxicity but can be seen in any cause of fibrosis, especially those associated with microvascular injury. It is likely that the associated arteriopathy and narrowing of the lumen contribute to development of fibrosis and atrophy after watershed infarcts within areas of ischemia. Local hypoxia leads to formation of free oxygen radicals, which promote cellular death by apoptosis. In addition, upregulation of transforming growth factor- β is considered an important etiologic factor in CNI toxicity.²⁸

Today, CNI nephrotoxicity is believed to be responsible for only a small proportion of graft failures. The Study of Long-term Deterioration of Kidney Allograft Function (DeKAF) demonstrated that 70% of patients with a primary or secondary histologic diagnosis of CNI nephrotoxicity had evidence of deposition of the complement split-product C4d and had DSA linking chronic graft failure to alloimmunity rather than CNI nephrotoxicity.^{31,32}

Recurrent and De Novo Glomerular Diseases

Recurrent glomerulonephritis (GN) is diagnosed by exclusion of donor-transmitted disease and de novo GN. Its relative importance for graft loss increases as graft survival lengthens.³³ Given that it is a relatively common occurrence and has implications for treatment and retransplantation, clinicians should carefully search for recurrence in patients with prior GN. Diagnosis and management of recurrent disease are discussed further in [Chapter 113](#).

Cardiovascular Risk Factors

The vasculopathy in chronic allograft injury resembles systemic vascular disease, raising the possibility that conventional cardiovascular (CV) disease risk factors may be implicated. Hypertension occurs in about 70% to 90% of transplant recipients and predisposes to graft failure.³⁴ Dyslipidemias, including raised total cholesterol, low-density lipoprotein cholesterol, and triglycerides and cigarette smoking, have also been associated with late graft failure. The management of CV risk factors in the transplant recipient is discussed in [Chapter 111](#).

PATHOGENESIS: IMMUNOLOGIC FACTORS

T-Cell–Mediated Rejection

Acute T-cell–mediated rejection, vascular or corticosteroid-resistant rejection, subclinical rejection, and late (chronic-active) T-cell–mediated rejection all contribute to the burden of chronic allograft injury. Early acute T-cell–mediated rejection (within 3 months of transplantation) that is adequately diagnosed and treated usually has no impact on long-term graft survival, but late chronic-active T-cell–mediated rejection in which serum creatinine does not return to baseline usually translates into IFTA and/or vascular damage, contributing to graft injury. Multiple and severe late acute T-cell–mediated rejection episodes are predictive of chronic graft dysfunction more than early cellular rejection and acute vascular rejection, which have stronger associations with acute graft loss.³⁵ There is some evidence from

TABLE 112.1 Updated 2019 Banff Classification Categories**Category 1: Normal Biopsy or Nonspecific Changes****Category 2: Antibody-Mediated Changes**

1. Active ABMR
2. Chronic active ABMR^b
3. Chronic ABMR
4. C4d staining without evidence of rejection

Active ABMR

All three features must be present for diagnosis.

1. Histologic evidence of acute tissue injury, including one or more of the following:
 - Microvascular inflammation ($g > 0$ and/or $ptc > 0$), in the absence of recurrent or de novo glomerulonephritis, although in the presence of acute TCMR, borderline infiltrate, or infection, $ptc \geq 1$ alone is not sufficient and g must be ≥ 1
 - Intimal or transmural arteritis ($v > 0$)^c
 - Acute TMA in the absence of any other cause
 - Acute tubular injury in the absence of any other apparent cause
2. Evidence of current/recent antibody interaction with vascular endothelium, including at least one of the following:
 - Linear C4d staining in peritubular capillaries (C4d2 or C4d3 by IF on frozen sections or C4d > 0 by IHC on paraffin sections)
 - At least moderate microvascular inflammation ($[g + ptc] \geq 2$), although in the presence of acute TCMR, borderline infiltrate, or infection; $ptc \geq 2$ alone is not sufficient, and g must be ≥ 1
 - Increased expression of gene transcripts in the biopsy tissue indicative of endothelial injury, if thoroughly validated
3. Serologic evidence of circulating donor-specific antibodies (DSA to HLA or other antigens). C4d staining or expression of validated transcripts/classifiers as noted previously in criterion 2 may substitute for DSA; however, thorough DSA testing, including testing for non-HLA antibodies if HLA antibody testing is negative, is strongly advised whenever criteria 1 and 2 are met.

Chronic Active ABMR

All three features must be present for diagnosis. As with active ABMR, biopsy samples showing histologic features plus evidence of current/recent antibody interaction with vascular endothelium or DSA, but not both, may be designated as suspicious, and it should be noted if the lesion is C4d positive or C4d negative, based on the criteria listed:

1. Histologic evidence of chronic tissue injury, including one or more of the following:
 - TG ($cg > 0$), if no evidence of chronic TMA; includes changes evident by EM only ($cg1a$; see Table 112.4)
 - Severe peritubular capillary basement membrane multilayering (requires EM)^f
 - Arterial intimal fibrosis of new onset, excluding other causes; leukocytes within the sclerotic intima favor chronic ABMR if there is no history of biopsy-proven TCMR with arterial involvement but are not required
2. Identical to criteria for active ABMR
3. Serologic evidence of DSAs (HLA or other antigens):
 - Biopsy results suspicious for ABMR on the basis of meeting criteria 1 and 2 should prompt expedited DSA testing

Chronic ABMR

All three features must be present for diagnosis:

1. $cg > 0$ and/or severe $ptcml$ ($ptcml1$)
2. Absence of criterion 2 of recurrent/recent antibody interaction with the endothelium
3. Prior documented diagnosis of active or chronic active ABMR and/or documented prior evidence of DSA

C4d Staining Without Evidence of Rejection

All four features must be present for diagnosis^d:

1. Linear C4d staining in peritubular capillaries (C4d2 or C4d3 by IF on frozen sections, or C4d > 0 by IHC on paraffin sections)
2. Criterion 1 for active or chronic active ABMR not met
3. No molecular evidence for ABMR as in criterion 2 for active and chronic active ABMR
4. No acute or chronic active TCMR, or borderline changes

Category 3: Borderline Changes Suspicious for Acute TCMR

- Foci of tubulitis ($t1$, $t2$, or $t3$) with mild interstitial inflammation ($i1$), or mild tubulitis ($t1$) with interstitial inflammation ($i2$, $i3$)
- No intimal arteritis ($v = 0$)

TABLE 112.1 Updated 2019 Banff Classification Categories—cont'd

Category 4: TCMR	Acute TCMR
1. Acute TCMR	IA. Significant interstitial inflammation (>25% of nonsclerotic cortical parenchyma, i2 or i3) and foci of moderate tubulitis (t2)
2. Chronic active TCMR	IB. Significant interstitial inflammation (>25% of nonsclerotic cortical parenchyma, i2 or i3) and foci of severe tubulitis (t3)
	IIA. Mild to moderate intimal arteritis (v1) with or without interstitial inflammation and tubulitis
	IIB. Severe intimal arteritis comprising > 25% of the luminal area (v2) with or without interstitial inflammation and tubulitis
	III. Transmural arteritis and/or arterial fibrinoid change and necrosis of medial smooth muscle cells with accompanying lymphocytic inflammation (v3)
	Chronic active TCMR
	IA. Interstitial inflammation involving >25% of the total cortex (ti score 2 or 3) and >25% of the sclerotic cortical parenchyma (i-IFTA score 2 or 3) with moderate tubulitis (t2) involving 1 or more tubules, not including severely atrophic tubules; other known causes of i-IFTA should be ruled out
	IB. Interstitial inflammation involving >25% of the total cortex (ti score 2 or 3) and >25% of the sclerotic cortical parenchyma (i-IFTA score 2 or 3) with severe tubulitis (t3) involving 1 or more tubules, not including severely atrophic tubules; other known causes of i-IFTA should be ruled out
	II. Chronic allograft arteriopathy (arterial intimal fibrosis with mononuclear cell inflammation in fibrosis and formation of neointima)
Category 5: Interstitial Fibrosis and Tubular Atrophy (IFTA)	Chronic allograft arteriopathy (arterial intimal fibrosis with mononuclear cell infiltration in fibrosis, formation of neointima); note that such lesions may represent chronic active ABMR and TCMR; the latter also may be manifest in the tubulointerstitial compartment
	I. Mild interstitial fibrosis and tubular atrophy (≤25% of cortical area)
	II. Moderate interstitial fibrosis and tubular atrophy (26%–50% of cortical area)
	III. Severe interstitial fibrosis and tubular atrophy (>50% of cortical area)
Category 6: Other Changes Not Considered Caused by Acute or Chronic Rejection	• BK virus nephropathy • Posttransplant lymphoproliferative disorders • Calcineurin inhibitor nephrotoxicity • Acute tubular injury • Recurrent disease • De novo glomerulopathy (other than transplant glomerulopathy) • Pyelonephritis • Drug-induced interstitial nephritis

^aThese arterial lesions may be indicative of ABMR, TCMR, or mixed ABMR/TCMR. The v lesions are scored in arteries only having a continuous media with two or more smooth muscle layers.

^bLesions of chronic, active ABMR can range from primarily active lesions with early TG evident only by EM to those with advanced TG and other chronic changes in addition to active microvascular inflammation. In the absence of evidence of current/recent antibody interaction with the endothelium (those features in the second section of Table 112.3), the term “active” should be omitted; in such cases, DSAs may be present at the time of biopsy or at any previous time after transplantation.

^cSeven or more layers in one cortical peritubular capillary and five or more in two additional capillaries, avoiding portions cut tangentially.

^dThe clinical significance of these findings may be quite different in grafts exposed to anti-blood group antibodies (ABO-incompatible allografts), in which they do not appear to be injurious to the graft and may represent accommodation; however, with anti-HLA antibodies, such lesions may progress to chronic ABMR and more outcome data are needed.

ABMR, Antibody-mediated rejection; *cg*, glomerular double contours; *DSA*, donor-specific antibody; *EM*, electron microscopy; *g*, glomerulitis; *i*, inflammation; *IF*, immunofluorescence; *IHC*, immunohistochemistry; *ptc*, peritubular capillaritis; *t*, tubulitis; *TCMR*, T-cell-mediated rejection; *TG*, transplant glomerulopathy; *TMA*, thrombotic microangiopathy; *v*, intimal arteritis.

From Loupy A, Haas M, Roufosse C, et al. The Banff 2019 Kidney meeting report (II): Updates on and clarification of criteria for T cell- and antibody-mediated rejection. *Am J Transplant*. 2020;20:2318–2331.

protocol biopsy studies that subclinical T-cell-mediated rejection may be an important factor contributing to chronic graft injury.^{36,37}

Besides the direct effects of T-cell-mediated rejection on graft function, acute T-cell-mediated rejection may be linked to the development of DSA and subsequent antibody-mediated rejection.³⁸

Antibody-Mediated Rejection

Chronic active antibody-mediated rejection is the leading cause of graft loss, together with death with a functioning graft, recurrent kidney disease, and IFTA of unknown origin.³⁹

For the diagnosis of antibody-mediated kidney graft rejection, histologic features of antibody-mediated rejection in the biopsy sample

(Figs. 112.2–112.4), together with evidence of current or recent antibody interaction with the vascular endothelium (see Fig. 112.3) and the detection of circulating DSA (either HLA or non-HLA), are required. Because C4d in peritubular capillaries is not detectable in more than 50% of patients with chronic antibody-mediated rejection, C4d-positivity is no longer a prerequisite for the diagnosis of antibody-mediated rejection.⁹ This finding, together with the observation that chronic antibody-mediated rejection may occur in the absence of complement activation (e.g., the occurrence of transplant glomerulopathy in patients during effective therapeutic complement inhibition by eculizumab), led to the hypothesis that complement-independent mechanisms such as direct endothelial cell activation and infiltration of natural

BOX 112.1 Main Factors Contributing to Late Graft Injury and Late Graft Loss

Donor Factors

- Deceased donor kidney (DBD or DCD)
- Older donor age, female donor
- Donor vascular disease
- Ischemia/reperfusion injury and long ischemia time
- Delayed graft function

Recipient Risks (Nonimmune)

- Obesity
- Urinary tract infection
- Transplant ureteral obstruction
- BK virus nephropathy
- Calcineurin inhibitor toxicity
- Recurrent kidney disease or de novo glomerulonephritis
- Hypertension, dyslipidemia, smoking
- Diabetes (preexisting or posttransplantation)

Recipient Risks (Alloimmune)

- Child or adolescent recipient
- Variable immunosuppressive medication trough concentrations from malabsorption or nonadherence
- HLA mismatches, presensitization status (donor-specific HLA alloantibodies)
- Acute rejection that is severe, corticosteroid resistant, vascular, antibody mediated, or late occurring
- Late de novo donor-specific antibodies and chronic (antibody-mediated) rejection

DBD, Donation after brain death; DCD, donation after cardiac death; HLA, human leukocyte antigen. Modified from Nankivell BJ, Kuypers DRJ. Diagnosis and prevention of chronic kidney allograft loss. *Lancet*. 2011;378:1428–1437.

killer (NK) cells or monocytes may contribute to chronic antibody-mediated allograft injury. C4d-negative antibody-mediated rejection shares the same histologic and ultrastructural features as C4d-positive rejection. To improve the diagnosis of antibody-mediated rejection, a molecular microscope strategy was adopted in the 2013 update of the Banff classification.⁴⁰ This means that in the absence of C4d positivity or microvascular inflammation, antibody interaction with the vascular endothelium is also suggested by detection of an increased expression of specific gene transcripts in the biopsy tissue indicative of endothelial injury (the “molecular microscope”), although this approach still needs further validation.

Unrecognized DSAs, if strongly reactive and complement-activating, can cause hyperacute rejection or accelerated antibody-mediated rejections in the early phase after kidney transplantation. Acute antibody-mediated rejection in the early phase occurs in about 1% to 6% of patients overall, but in as many as 21% to 55% of patients who had detectable DSAs already before transplantation and who received desensitization therapy.^{41–43} Weakly positive DSAs have been associated with rather subtle types of graft damage, often leading to delayed graft function.² Early damage can lead to chronic rejection, most probably because the structure of the endothelium is no longer intact and new antigenic epitopes are expressed on the surface of transplanted tissue. During later phases after transplantation, insufficient immunosuppression can facilitate the development of de novo DSAs against these new epitopes and result in chronic antibody-mediated rejection and failure of the transplanted organ. De novo DSAs, but to a lesser extent also persistence or reemergence of DSAs that were

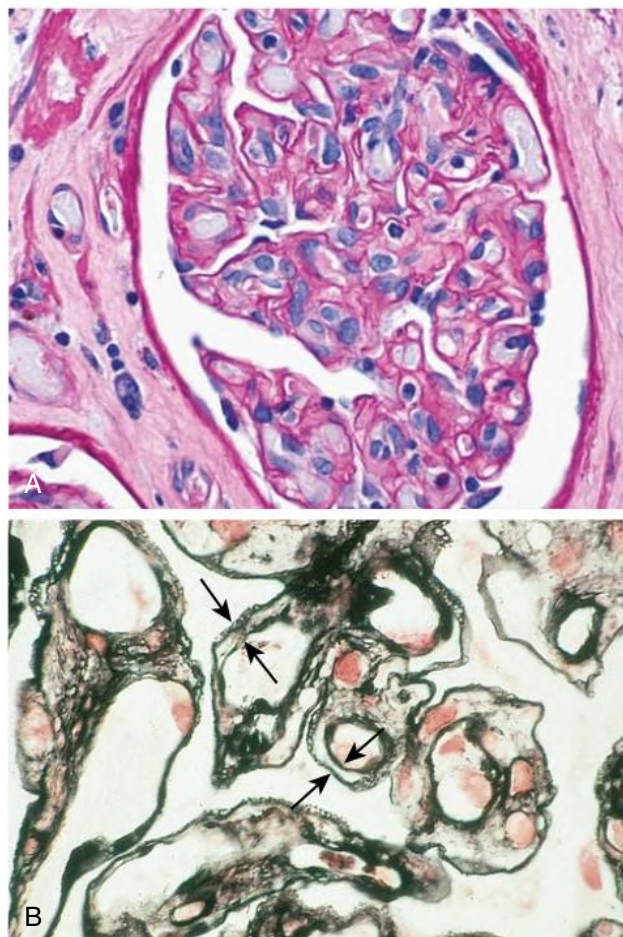


Fig. 112.2 Transplant Glomerulopathy: Light Microscopy. (A) Note the mesangial expansion and thickening of the glomerular basement membrane (GBM). (B) Note the GBM reduplication in capillary loops (between arrows).

detectable before transplantation, are associated with poor allograft outcome.

Chronic active antibody-mediated rejection is found more frequently in patients who are nonadherent to immunosuppressive medication or in whom immunosuppression was reduced for other reasons, such as conversion to CNI-free or steroid-free immunosuppressive protocols, recurrent infection, or malignancy.^{44–46} In these patients, antibody-mediated lesions and T-cell-mediated lesions are often found in kidney graft biopsy samples at the same time. Additional risk factors for the development of de novo DSAs and antibody-mediated rejection are HLA class II DR mismatches between donor and recipient, prior cellular rejection episodes, and younger recipient age. In many patients with late antibody-mediated graft loss, even when HLA class I alloantibodies are detectable, HLA class II de novo DSAs are considered to be mainly responsible for rejection. In a recent study on the evolution of HLA alloantibodies after transplantation, de novo DSAs appeared at a mean of 4.6 years after transplantation, and the prevalence of de novo DSAs after 10 years was 20% in those adherent to the immunosuppressive regimen, compared with 60% in nonadherent graft recipients.⁴⁶ In another study, DSAs were found in 37% of patients who had an indication biopsy 7 days to 31 years after transplant.⁷ De novo DSAs that were directed against HLA class II antigens had an especially strong association with strongly impaired graft survival. A recent development for the detection of harmful DSAs is the introduction of solid-phase assays into a clinical routine, which

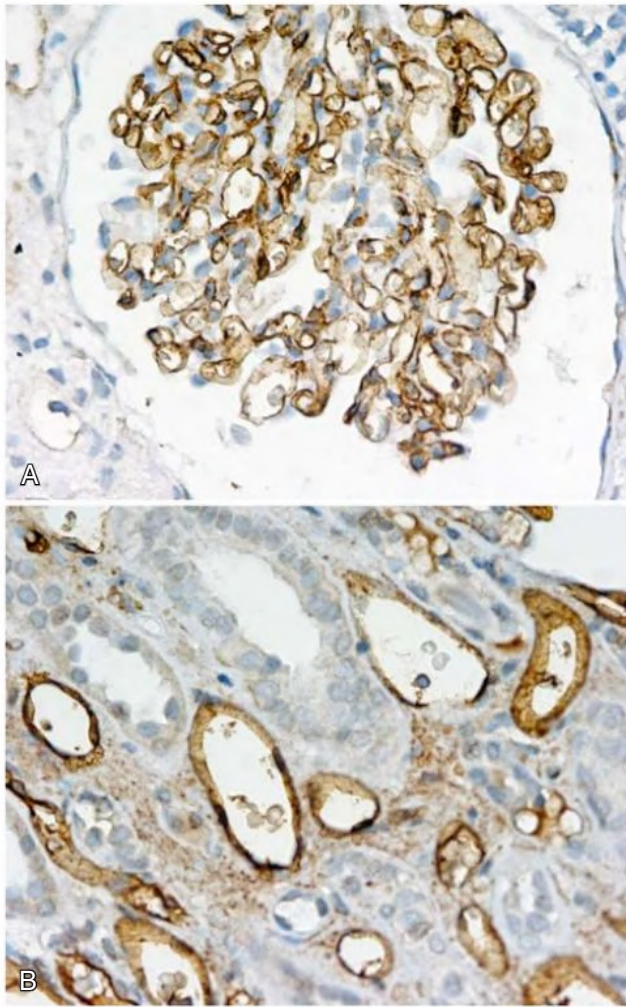


Fig. 112.3 C4d Immunoperoxidase Staining in Association With Chronic Antibody-Mediated Rejection. Staining of glomerular capillaries (A) and circumferential staining of peritubular capillaries (B) are typical.

enables the distinction of complement-binding (C1q assay) HLA antibodies from antibodies that do not bind complement. In a cohort of more than 1000 patients, it was recently demonstrated that the post-transplant occurrence of complement-binding DSA is associated with adverse events. The 5-year graft survival in patients with complement-binding antibodies was 54% compared with 93% in patients with DSAs that were not able to bind complement. The higher risk for graft loss in patients with complement-binding DSAs was attributable to a higher risk for chronic antibody-mediated rejection.⁴⁷ Recent preliminary data also suggest that DSAs of different immunoglobulin G subclasses may be associated with different distinct injury phenotypes in kidney allograft biopsies.

Other antibodies that may contribute to chronic antibody-mediated rejection are MHC class I-related chain A antibodies, angiotensin II type 1 receptor-activating antibodies, and other antiendothelial cell antibodies.^{48,49}

Insufficient Immunosuppression and Nonadherence to Medication

The recognition that HLA alloantibodies are responsible for a great proportion of late graft losses^{37,50} has focused attention on insufficient immunosuppression and nonadherence to immunosuppressive medication (Fig. 112.5).^{44,51}

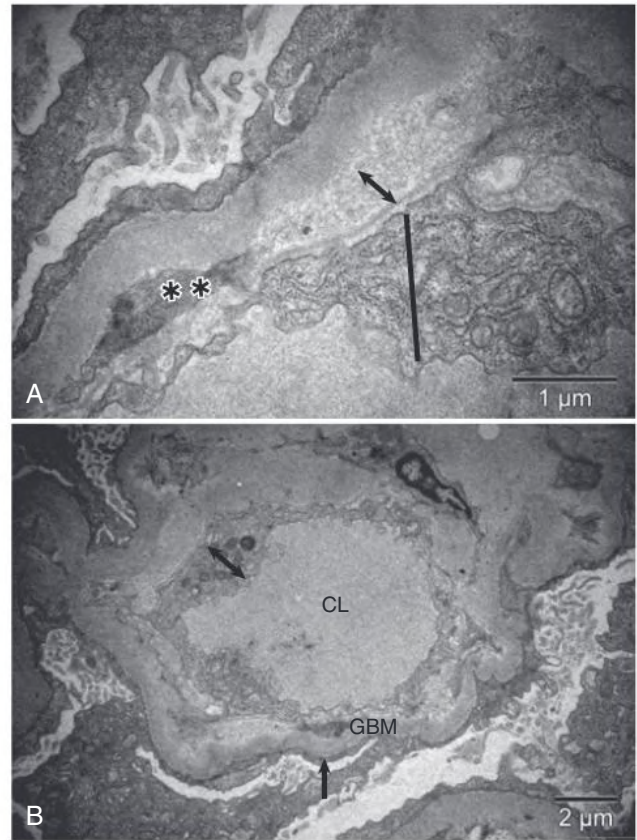


Fig. 112.4 Transplant Glomerulopathy: Electron Microscopy. (A) Section of a glomerular capillary loop ($\times 24,000$) demonstrating mesangial interposition (*asterisks*), subendothelial expansion and new lamina densa (*double arrow*), and endothelial hypertrophy (*line*). (B) A complete glomerular capillary loop (CL) with glomerular basement membrane (GBM) thickening (*single arrow*), endothelial hypertrophy (*double arrow*), expanded subendothelial space, and new lamina densa (*single arrow*). Apparent GBM duplication is a result of mesangial interposition and formation of new lamina densa.

A retrospective analysis of more than 25,000 kidney transplant recipients showed that reduction or discontinuation of cyclosporine, tacrolimus, or mycophenolate mofetil after the first posttransplant year in patients with good graft function was associated with significantly reduced subsequent kidney graft survival.⁴⁴ In a recent study in which 64% of graft losses in a cohort with indication biopsies were attributable to (antibody-mediated) rejection, about half of the patients with rejection-associated allograft loss were identified as nonadherent.⁵¹

Young adults who are in transition from pediatric to adult nephrology care are at high risk for nonadherence. Other risk factors are previous nonadherence, psychiatric disorders, substance abuse, insufficient socioeconomic support, and adverse effects from immunosuppressive medication. Insufficient immunosuppression may also occur during immunosuppressive minimization (tapering) or CNI-avoidance trials. In a recent study, 14 of 61 patients (23%) who were converted from cyclosporine to everolimus at 3 to 4.5 months after transplantation developed DSA, compared with only 7 of 65 patients (11%) who continued on cyclosporine.⁴¹ Eight patients on everolimus but only two patients on cyclosporine developed antibody-mediated rejection. Therefore, such patients should be rigorously evaluated for alloantibodies and antibody-mediated allograft injury according to Transplantation Society Guidelines (see later).⁵²

Insufficient Immunosuppression Contributes to Late Graft Loss

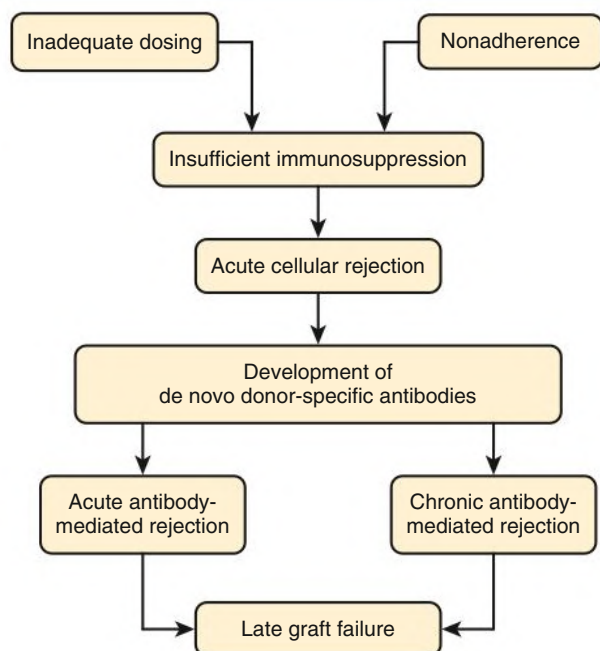


Fig. 112.5 Insufficient immunosuppression contributes to late graft loss.

CLINICAL MANIFESTATIONS

Routine laboratory evaluation of graft function does not facilitate the early diagnosis of chronic allograft injury. A rise in serum creatinine is a late marker of allograft dysfunction. Reduction in the glomerular filtration rate is usually estimated by equations such as the Modification of Diet in Renal Disease (MDRD) formula or the chronic kidney disease epidemiology collaboration (CKD-EPI) formula. However, substantial nephron loss may occur before a rise in serum creatinine becomes evident.

Persistent or worsening proteinuria is also a late sign of chronic allograft injury. Proteinuria greater than 1 g/day typically occurs with recurrent or de novo GN or with severe chronic antibody-mediated changes such as transplant glomerulopathy (see Figs. 112.2–112.4). A definite diagnosis of antibody-mediated rejection requires transplant kidney biopsy together with serologic testing for alloantibodies and viral disease such as BK virus or cytomegalovirus (CMV) infection. So far, no noninvasive tests can reliably diagnose antibody-mediated rejection. The evolution of antibody-mediated graft damage from the binding of DSAs to graft loss is given in Fig. 112.6.

Recently a risk prediction score for long-term kidney allograft failure has been developed and validated. The “iBox” combines functional, histologic, and immunologic allograft parameters together with HLA antibody profiling and showed good performance in predicting the risk of long-term kidney allograft failure.⁵³

The Evolution of Antibody-Mediated Graft Damage

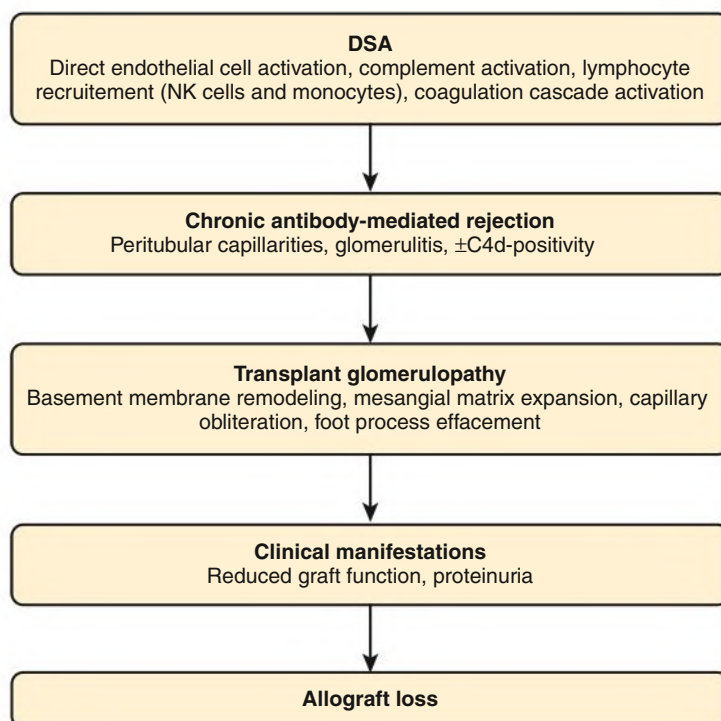


Fig. 112.6 Evolution of antibody-mediated graft damage from the binding of donor-specific antibodies (DSA) to graft loss. *NK*, Natural killer.

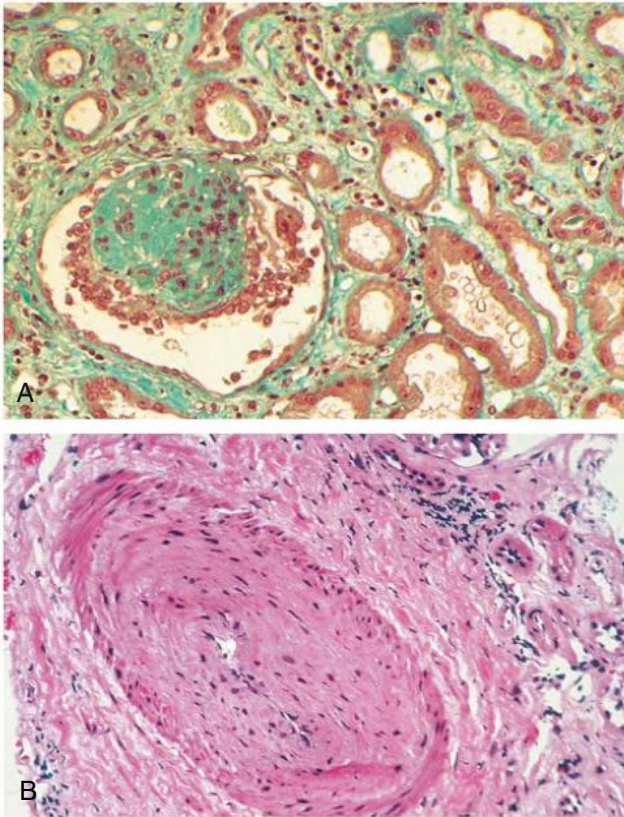


Fig. 112.7 Histologic Features of Chronic Allograft Injury and Interstitial Fibrosis–Tubular Atrophy With No Evidence of Specific Cause. (A) Interstitial fibrosis and glomerulosclerosis. (Trichrome stain.) (B) Fibrointimal proliferation in an intrarenal artery. (PAS stain.)

PATHOLOGY

Chronic IFTA (Fig. 112.7) represents the final pathway of nephron injury and is not specific for any graft disease. IFTA is found in most late graft biopsies, whether a result of early graft damage (e.g., tubular injury from ischemia reperfusion injury, overt or subclinical rejection, or even preexisting disease of the donor) or later graft damage (secondary to CNI toxicity, hypertension and hyperlipidemia, recurrent glomerular disease, or immune-mediated injury) or of unknown origin (IFTA).

Chronic antibody-mediated rejection is characterized by structural remodeling of glomerular basement membranes (GBMs; so-called transplant glomerulopathy) and similar changes in the peritubular capillaries together with IFTA and fibrous thickening of arteries (see Figs. 112.2–112.4). Transplant glomerulopathy is characterized by duplication of glomerular capillary basement membranes and mesangial matrix expansion, in the absence of immune deposits. In addition, deposition of C4d in peritubular and glomerular capillaries can sometimes occur (see Fig. 112.3). Clinicopathologic studies suggest that transplant glomerulopathy is a manifestation of capillary injury occurring in conjunction with interstitial, peritubular capillary, and glomerular inflammation, although it also may occur independently of IFTA or transplant arteriopathy.⁵⁴ On electron microscopy (EM), there is expansion of the subendothelial space with deposition of flocculent or fibrillary material, interposition of mesangial cell cytoplasm in the lamina densa, and mesangial matrix expansion (see Fig. 112.4). EM may be used to diagnose transplant glomerulopathy at its earliest stages. Ultrastructural abnormalities characteristic of endothelial

activation occur long before GBM duplication and graft dysfunction are evident, implying that endothelial injury is the initial insult that resulted in GBM remodeling.⁵⁵ An association between transplant glomerulopathy and peritubular capillary basement membrane duplication has been well described, suggesting that the process resulting in transplant glomerulopathy involves the entire glomerular and peritubular capillary beds.^{56,57} It is currently debated whether transplant glomerulopathy represents scarring that is refractory to treatment.

DIAGNOSIS AND DIFFERENTIAL DIAGNOSIS

When chronic allograft injury has been identified by a gradual rise in serum creatinine and/or by new or worsening proteinuria, ultrasound should be performed to exclude ureteral or vascular problems and assess the Doppler resistive index. The Doppler resistive index gives an estimate of intrarenal (but also systemic) compliance and is most useful in longitudinal studies in the same patient to assess progressive graft vascular injury.

In addition, testing for BK virus in plasma is also advised during deteriorating graft function and for all kidney transplant recipients at 3-month intervals during the first 2 years after transplantation. Circulating HLA alloantibodies also should be documented during graft dysfunction and may further guide diagnostics and therapy. Many centers now perform routine HLA alloantibody screening in stable graft recipients at different time points after transplantation to facilitate early diagnosis of antibody-mediated rejection. According to the 2013 The Transplantation Society Consensus Guidelines, non-sensitized first kidney transplant recipients should be tested for DSAs at least once 3 to 12 months after transplantation.⁵² In presensitized patients, more frequent testing is recommended. Although there is consensus that HLA alloantibodies are responsible for a significant proportion of late graft losses and that HLA antibodies in the context of deteriorating graft function are harmful, the significance of HLA alloantibodies that are detected solely during routine testing is uncertain.^{58,59}

Allograft biopsy is required to rule out a specific allograft pathology together with specific staining, such as C4d, for detection of antibody-mediated allograft injury in all patients and SV40 if there is suspicion for BK virus nephropathy. Other specific examinations of the biopsy specimen for immunoglobulins, complement components, or EM may in addition be useful to rule out recurrence of primary kidney disease or de novo GN. The role of protocol biopsies in the early detection of chronic allograft injury remains controversial.

Diagnosing allograft injury from blood samples represents a new area of research. For example, donor-derived cell-free DNA (dd-cfDNA) comes from the transplanted organ and is exogenous to the respective kidney transplant recipient. An increase of dd-cfDNA levels in the peripheral blood seems to occur even before the serum creatinine level starts rising, which may enable early diagnosis of transplant injury and ultimately avoid premature graft loss.⁶⁰ However, randomized controlled trials are still lacking and no firm recommendations can be made.⁶¹

Because the etiology of chronic allograft dysfunction is multifactorial, any specific diagnosis is made by combination of a biopsy with a review of the patient's history to identify important etiologic factors, such as preexisting donor disease, prior rejection, and high-titer anti-HLA antibodies, as well as issues not related to alloimmunity, such as CNI nephrotoxicity, de novo GN, or recurrent kidney disease. Chronic allograft dysfunction is common and can have many causes. If it is left untreated, it can ultimately result in kidney injury, which, in turn, will heal by scarring and interstitial fibrosis and progress to ESKD.

Prevention of Antibody-Mediated Allograft Injury

In the absence of established treatments of circulating DSAs and chronic antibody-mediated rejection, only prevention of antibody-mediated allograft injury is effective in preventing future graft loss. Once harmful HLA antibodies such as C1q-binding DSA are detectable, it becomes very difficult to avoid allograft injury and decline of kidney function. Preventive measures include avoidance of sensitization by limitation of blood transfusions and poor HLA matching during first transplantations.^{62,63} The introduction of epoetin led to a dramatic decrease of recipient sensitization by a significant reduction in blood transfusions.⁶⁴ Sensitization should especially be avoided in young patients who may require retransplantation during their lifetime. New measures, such as matching for antibody epitopes in addition to whole HLA alleles, may further help preventing sensitization.

When the patient is already sensitized, antibody-mediated allograft injury can best be prevented by transplantation of the patient with an organ against which the recipient has not developed HLA alloantibodies. This may be achieved by inclusion of patients in special programs such as Eurotransplant Acceptable Mismatch Program or by kidney-paired donation.^{64,65} However, successful transplantation in patients with preexisting HLA alloantibodies usually requires desensitization in combination with other measures.⁶⁶

After transplantation, chronic antibody-mediated allograft rejection is best prevented by conventional triple-drug therapy including a CNI (preferably tacrolimus), an antiproliferative agent (preferably mycophenolic acid-based therapy), and corticosteroids. Induction therapy, with an interleukin-2 receptor antagonist or antithymocyte globulin, should be used to prevent early acute cellular rejection.

Insufficient immunosuppression and nonadherence may lead, unless a patient is prone to tolerance, to cellular rejection and then, via development of de novo DSA, to antibody-mediated rejection and graft loss. Precise knowledge of the patient's alloantibody status before and after transplantation is a prerequisite for early diagnosis of allograft injury and early and targeted treatment to prevent antibody-mediated rejection and ensure long-term graft survival. HLA alloantibodies should be tested at least once after transplantation in all patients. In immunologically high-risk patients, desensitized patients, patients with suspected rejection, and during therapy of antibody-mediated rejection, antibodies need to be monitored more frequently with the goal of recognizing allograft injury in its early stages and preventing its progression to chronic rejection.⁵² Additional patient groups who may benefit from HLA alloantibody monitoring are patients who receive reduction, withdrawal, or change of immunosuppressive drugs, especially in the context of CNI- or steroid-free regimens. Protocol biopsies may further help in guiding posttransplant therapy at least in high-risk recipients, although definitive proof is lacking.

Strategies for the prevention of chronic allograft injury are summarized in [Box 112.2](#).

Treatment of Chronic Allograft Injury

Although reduction of blood pressure to less than 130/80 mm Hg is believed to be beneficial, the preferred agent for blood pressure reduction remains controversial; in particular the use of angiotensin-converting enzyme (ACE) inhibitors and angiotensin receptor

BOX 112.2 Strategies for the Prevention of Chronic Allograft Injury

- Avoid sensitization in future organ recipients (e.g., avoid blood transfusions, avoid poor HLA matching during first transplantation).
- Obtain complete donor and recipient typing and precise knowledge on HLA alloantibodies at time of transplantation.
- Minimize ischemia/reperfusion injury (e.g., short cold ischemia time).
- Avoid insufficient immunosuppression.
- Perform prophylaxis for CMV (first 3–6 months).
- Test for BK virus after transplantation.
- Monitor donor-specific HLA alloantibodies after transplantation (at least in presensitized patients or during drug minimization).
- Perform protocol biopsies to detect subclinical rejection (at least in presensitized patients or during drug minimization).
- Review for nonadherence.

CMV, Cytomegalovirus; HLA, human leukocyte antigen.

BOX 112.3 Treatment of Chronic Allograft Injury

Treatment (Nonimmune)

- Treat hypertension (consider ACE inhibitors or AT₁ receptor antagonists)
- Lifestyle modification (stop smoking, control lipids)
- Control diabetes
- Prevention and treatment of urinary tract infection
- Target CNI toxicity by reduction or replacement of CNI (caveat: risk for insufficient immunosuppression)

Treatment (Alloimmune)

- Diagnose and treat early acute cellular rejection by steroids or antithymocyte globulin
- Diagnose and treat acute antibody-mediated rejection by means such as plasmapheresis or immunoadsorption ± anti-CD20 therapy
- Diagnose and treat chronic antibody-mediated rejection by means such as intravenous immunoglobulins ± anti-CD20 therapy
- Consider proteasome inhibitor, CD38 inhibitors, IL-6 receptor inhibitor, complement blockade, or the cysteine protease imlifidase (no clear evidence).

ACE, Angiotensin-converting enzyme; AT₁, angiotensin 1 receptor; CNI, calcineurin inhibitor.

blockers (ARBs) has not uniformly shown benefit.^{67,68} Nor have there been specific studies confirming the benefits of reducing proteinuria in chronic allograft injury, although by extrapolation for other progressive proteinuric diseases, there is strong a priori evidence for the benefits of renin-angiotensin system blockade with dietary salt restriction.

A beneficial effect of treating dyslipidemia with statins in the transplant recipient also remains controversial,^{69,70} and the adverse effects of statins such as rhabdomyolysis when used with CNIs must be considered. After a specific diagnosis had been established, targeted treatments can be initiated. These are summarized in [Box 112.3](#).

SELF-ASSESSMENT QUESTIONS

1. Which one of the following is believed to be mainly responsible for late kidney graft loss?
 - A. Hypertension and hyperlipidemia
 - B. New-onset diabetes after transplantation
 - C. Recurrence of primary disease
 - D. HLA alloantibodies
 - E. Calcineurin inhibitors
2. Chronic antibody-mediated rejection:
 1. may be C4d negative in graft biopsy samples.
 2. has specific histopathologic features.
 3. is found more often in patients nonadherent to immunosuppressive medication.
 4. can be mediated only by HLA alloantibodies.
 5. is targeted by reduction of immunosuppression.
 - A. 1 and 2 are correct
 - B. 1, 2, 3, and 4 are correct
 - C. 1, 2, and 3 are correct
 - D. All are correct
 - E. No answer is correct
3. Clinical manifestations of chronic antibody-mediated rejection are:
 1. a rise in serum creatinine.
 2. a rise in protein excretion.
 3. specific for chronic antibody-mediated rejection.
 4. often late signs that occur after significant allograft damage has already occurred.
 - A. Only 1 is correct
 - B. 1 and 2 are correct
 - C. All are correct
 - D. 1, 2, and 3 are correct
 - E. 1, 2, and 4 are correct

REFERENCES

- Ekberg H, Tedesco-Silva H, Demirbas A, et al. Reduced exposure to calcineurin inhibitors in renal transplantation. *N Engl J Med*. 2007;357:2562–2575.
- Süsal C, Döhler B, Sadeghi M, et al. HLA antibodies and the occurrence of early adverse events in the modern era of transplantation: a collaborative transplant study report. *Transplantation*. 2009;87:1367–1371.
- Dragun D, Müller DN, Bräsen JH, et al. Angiotensin II type 1-receptor activating antibodies in renal-allograft rejection. *N Engl J Med*. 2005;352(6):558–569.
- Reindl-Schwaighofer R, Heinzel A, Kainz A, et al. Contribution of non-HLA incompatibility between donor and recipient to kidney allograft survival: genome-wide analysis in a prospective cohort. *Lancet*. 2019;393(10174):910–917.
- Lefaucheur C, Louis K, Philippe A, Loupy A, Coates PT. The emerging field of non-human leukocyte antigen antibodies in transplant medicine and beyond. *Kidney Int*. 2021;100(4):787–798.
- Nankivell BJ, Borrows RJ, Fung CL, et al. The natural history of chronic allograft nephropathy. *N Engl J Med*. 2003;349:2326–2333.
- Hidalgo LG, Campbell PM, Sis B, et al. De novo donor-specific antibody at the time of kidney transplant biopsy associates with microvascular pathology and late graft failure. *Am J Transplant*. 2009;9:2532–2541.
- Loupy A, Haas M, Roufosse C, et al. The Banff 2019 Kidney meeting report (I): Updates on and clarification of criteria for T cell- and antibody-mediated rejection. *Am J Transplant*. 2020;20:2318–2331.
- Mengel M, Sis B, Haas M, et al. Banff 2011 meeting report: new concepts in antibody-mediated rejection. *Am J Transplant*. 2012;12:563–570.
- Bos EM, Leuvenink HG, van Goor H, Ploeg RJ. Kidney grafts from brain dead donors: inferior quality or opportunity for improvement? *Kidney Int*. 2007;72:797–805.
- Young A, Kim SJ, Speechley MR, et al. Accepting kidneys from older living donors: impact on transplant recipient outcomes. *Am J Transplant*. 2011;11:743–750.
- Kainz A, Perco P, Mayer B, et al. Gene-expression profiles and age of donor kidney biopsies obtained before transplantation distinguish medium term graft function. *Transplantation*. 2007;83:1048–1054.
- Halloran PF, Melk A, Barth C. Rethinking chronic allograft nephropathy: the concept of accelerated senescence. *J Am Soc Nephrol*. 1999;10:167–181.
- Saxena AB, Busque S, Arjane P, et al. Preoperative renal volumes as a predictor of graft function in living donor transplantation. *Am J Kidney Dis*. 2004;44:877–885.
- Gratwohl A, Dohler B, Stern M, Opelz G. H-y as a minor histocompatibility antigen in kidney transplantation: a retrospective cohort study. *Lancet*. 2008;372:49–53.
- Zeier M, Dohler B, Opelz G, Ritz E. The effect of donor gender on graft survival. *J Am Soc Nephrol*. 2002;13:2570–2576.
- Schwarz A, Mengel M, Gwinner W, et al. Risk factors for chronic allograft nephropathy after renal transplantation: a protocol biopsy study. *Kidney Int*. 2005;67:341–348.
- Kim IK, Bedi DS, Denecke C, et al. Impact of innate and adaptive immunity on rejection and tolerance. *Transplantation*. 2008;86:889–894.
- Schnuelle P, Gottmann U, Hoeger S, et al. Effects of donor pretreatment with dopamine on graft function after kidney transplantation: a randomized controlled trial. *JAMA*. 2009;302:1067–1075.
- Schnuelle P, Schmitt WH, Weiss C, et al. Effects of dopamine donor pretreatment on graft survival after kidney transplantation: a randomized trial. *Clin J Am Soc Nephrol*. 2017;12:493–501.
- Niemann CU, Feiner J, Swain S, et al. Therapeutic hypothermia in deceased organ donors and kidney-graft function. *N Engl J Med*. 2015;373:405–414.
- Kuypers DR, Chapman JR, O'Connell PJ, et al. Predictors of renal transplant histology at three months. *Transplantation*. 1999;67:1222–1230.
- Nickeleit V, Singh HK, Randhawa P, et al. The Banff Working Group classification of definitive polyomavirus nephropathy: morphologic definitions and clinical correlations. *J Am Soc Nephrol*. 2018;29(2):680–693.
- Hirsch HH, Randhawa P. Bk polyomavirus in solid organ transplantation. *Am J Transplant*. 2013;13(suppl 4):179–188.
- Nankivell BJ, Borrows RJ, Fung CL, et al. Calcineurin inhibitor nephrotoxicity: longitudinal assessment by protocol histology. *Transplantation*. 2004;78:557–565.
- Benigni A, Bruzzi I, Mister M, et al. Nature and mediators of renal lesions in kidney transplant patients given cyclosporine for more than one year. *Kidney Int*. 1999;55:674–685.
- Antonovych TT, Sabnis SG, Austin HA, et al. Cyclosporine a-induced arteriopathy. *Transplant Proc*. 1988;20:951–958.
- Austin 3rd HA, Palestine AG, Sabnis SG, et al. Evolution of cyclosporin nephrotoxicity in patients treated for autoimmune uveitis. *Am J Nephrol*. 1989;9:392–402.
- Mengel M, Mihatsch M, Halloran PF. Histological characteristics of calcineurin inhibitor toxicity—there is no such thing as specificity. *Am J Transplant*. 2011;11:2549–2550.
- Gaston RS, Cecka JM, Kasiske BL, et al. Evidence for antibody-mediated injury as a major determinant of late kidney allograft failure. *Transplantation*. 2010;90:68–74.
- Briganti EM, Russ GR, McNeil JJ, et al. Risk of renal allograft loss from recurrent glomerulonephritis. *N Engl J Med*. 2002;347:103–109.
- Opelz G, Wujciak T, Ritz E. Association of chronic kidney graft failure with recipient blood pressure. Collaborative transplant study. *Kidney Int*. 1998;53:217–222.
- Kasiske BL, Kalil RS, Lee HS, Rao KV. Histopathologic findings associated with a chronic, progressive decline in renal allograft function. *Kidney Int*. 1991;40:514–524.
- Nankivell BJ, Borrows RJ, Fung CL, et al. Natural history, risk factors, and impact of subclinical rejection in kidney transplantation. *Transplantation*. 2004;78:242–249.
- Rush D, Nickerson P, Gough J, et al. Beneficial effects of treatment of early subclinical rejection: a randomized study. *J Am Soc Nephrol*. 1998;9:2129–2134.
- Schaefer SM, Susal C, Sommerer C, et al. Current pharmacotherapeutical options for the prevention of kidney transplant rejection. *Expert Opin Pharmacother*. 2013;14:1029–1041.
- Einicke G, Sis B, Reeve J, et al. Antibody-mediated microcirculation injury is the major cause of late kidney transplant failure. *Am J Transplant*. 2009;9:2520–2531.
- Haas M, Sis B, Racusen LC, et al. Banff 2013 meeting report: inclusion of c4d-negative antibody-mediated rejection and antibody-associated arterial lesions. *Am J Transplant*. 2014;14:272–283.
- Amico P, Honger G, Mayr M, et al. Clinical relevance of pretransplant donor-specific HLA antibodies detected by single-antigen flow-beads. *Transplantation*. 2009;87:1681–1688.
- Gloor JM, Winters JL, Cornell LD, et al. Baseline donor-specific antibody levels and outcomes in positive crossmatch kidney transplantation. *Am J Transplant*. 2010;10:582–589.
- Lefaucheur C, Loupy A, Hill GS, et al. Preexisting donor-specific HLA antibodies predict outcome in kidney transplantation. *J Am Soc Nephrol*. 2010;21:1398–1406.
- Opelz G, Dohler B. Effect on kidney graft survival of reducing or discontinuing maintenance immunosuppression after the first year posttransplant. *Transplantation*. 2008;86:371–376.
- Liefeldt L, Brakemeier S, Glander P, et al. Donor-specific HLA antibodies in a cohort comparing everolimus with cyclosporine after kidney transplantation. *Am J Transplant*. 2012;12:1192–1198.
- Wiebe C, Gibson IW, Blydt-Hansen TD, et al. Evolution and clinical pathologic correlations of de novo donor-specific HLA antibody post kidney transplant. *Am J Transplant*. 2012;12:1157–1167.
- Loupy A, Lefaucheur C, Vernerey D, et al. Complement-binding anti-HLA antibodies and kidney-allograft survival. *N Engl J Med*. 2013;369:1215–1226.
- Zou Y, Stastny P, Süsal C, et al. Antibodies against mica antigens and kidney-transplant rejection. *N Engl J Med*. 2007;357:1293–1300.
- Dragun D, Muller DN, Bräsen JH, et al. Angiotensin II type 1-receptor activating antibodies in renal-allograft rejection. *N Engl J Med*. 2005;352:558–569.
- Terasaki PI. Humoral theory of transplantation. *Am J Transplant*. 2003;3:665–673.

49. Sellares J, de Freitas DG, Mengel M, et al. Understanding the causes of kidney transplant failure: the dominant role of antibody-mediated rejection and nonadherence. *Am J Transplant*. 2011.
50. Tait BD, Süsal C, Gebel HM, et al. Consensus guidelines on the testing and clinical management issues associated with HLA and non-HLA antibodies in transplantation. *Transplantation*. 2013;95:19–47.
51. Loupy A, Aubert O, Orandi BJ, et al. Prediction system for risk of allograft loss in patients receiving kidney transplants: international derivation and validation study. *BMJ*. 2019;366:l4923.
52. Gloor JM, Sethi S, Stegall MD, et al. Transplant glomerulopathy: subclinical incidence and association with alloantibody. *Am J Transplant*. 2007;7:2124–2132.
53. Wavamunno MD, O'Connell PJ, Vitalone M, et al. Transplant glomerulopathy: ultrastructural abnormalities occur early in longitudinal analysis of protocol biopsies. *Am J Transplant*. 2007;7:2757–2768.
54. Ivanyi B. Transplant capillaropathy and transplant glomerulopathy: ultrastructural markers of chronic renal allograft rejection. *Nephrol Dial Transplant*. 2003;18:655–660.
55. Regele H, Bohmig GA, Habicht A, et al. Capillary deposition of complement split product c4d in renal allografts is associated with basement membrane injury in peritubular and glomerular capillaries: a contribution of humoral immunity to chronic allograft rejection. *J Am Soc Nephrol*. 2002;13:2371–2380.
56. Süsal C, Morath C. Current approaches to the management of highly sensitized kidney transplant patients. *Tissue Antigens*. 2011;77:177–186.
57. Morath C, Opelz G, Zeier M, Susal C. Prevention of antibody-mediated kidney transplant rejection. *Transpl Int*. 2012;25:633–645.
58. Oellerich M, Shipkova M, Asendorf T, et al. Absolute quantification of donor-derived cell-free DNA as a marker of rejection and graft injury in kidney transplantation: results from a prospective observational study. *Am J Transplant*. 2019;19(11):3087–3099.
59. Filippone EJ, Farber JL. The monitoring of donor-derived cell-free DNA in kidney transplantation. *Transplantation*. 2021;105(3):509–516.
60. Meier-Kriesche HU, Scornik JC, Susskind B, et al. A lifetime versus a graft life approach redefines the importance of HLA matching in kidney transplant patients. *Transplantation*. 2009;88:23–29.
61. Vella JP, O'Neill D, Atkins N, et al. Sensitization to human leukocyte antigen before and after the introduction of erythropoietin. *Nephrol Dial Transplant*. 1998;13:2027–2032.
62. Claas FH, Rahmel A, Doxiadis II. Enhanced kidney allocation to highly sensitized patients by the acceptable mismatch program. *Transplantation*. 2009;88:447–452.
63. Montgomery RA. Renal transplantation across HLA and ABO antibody barriers: Integrating paired donation into desensitization protocols. *Am J Transplant*. 2010;10:449–457.
64. Morath C, Beimler J, Opelz G, et al. An integrative approach for the transplantation of high-risk sensitized patients. *Transplantation*. 2010;90:645–653.
65. Opelz G, Zeier M, Laux G, et al. No improvement of patient or graft survival in transplant recipients treated with angiotensin-converting enzyme inhibitors or angiotensin II type 1 receptor blockers: a collaborative transplant study report. *J Am Soc Nephrol*. 2006;17:3257–3262.
66. Heinze G, Mitterbauer C, Regele H, et al. Angiotensin-converting enzyme inhibitor or angiotensin ii type 1 receptor antagonist therapy is associated with prolonged patient and graft survival after renal transplantation. *J Am Soc Nephrol*. 2006;17:889–899.
67. Wiesbauer F, Heinze G, Mitterbauer C, et al. Statin use is associated with prolonged survival of renal transplant recipients. *J Am Soc Nephrol*. 2008;19:2211–2218.
68. Holdaas H, Fellstrom B, Jardine AG, et al. Effect of fluvastatin on cardiac outcomes in renal transplant recipients: a multicentre, randomised, placebo-controlled trial. *Lancet*. 2003;361:2024–2031.
69. Seron D, Oppenheimer F, Pallardo LM, et al. Fluvastatin in the prevention of renal transplant vasculopathy: results of a prospective, randomized, double-blind, placebo-controlled trial. *Transplantation*. 2008;86:82–87.
70. Nankivell BJ, Kuypers DRJ. Diagnosis and prevention of chronic kidney allograft loss. *Lancet*. 2011;378:1428–1437.